Linear Algebra | Quiz for §4.6 | October 30, 2026

Name: _____ R. Hammack Score: _____

Directions: Show and explain all work. Put phones away. No calculators, etc.

This quiz concerns the matrix $A = \begin{bmatrix} 3 & 3 & 15 & 11 \\ 1 & -3 & 1 & 1 \\ 2 & 3 & 11 & 11 \end{bmatrix}$

1. Find a basis for the row space of A.

2. Find a basis for the column space of A. (You may use your work from part 1 above.)

3. Find the dimension of the null space of A, that is, find nulity(A).
   (You may do this either by finding a basis for Null(A), or by using the Rank-Nullity Theorem.)